

Q vs DQN - Stat Tests

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1 Imports

```
[3]: import numpy as np
import pandas as pd
from scipy.stats import permutation_test
```

2 Data Loading

```
[4]: models = [
    "logit", # row 0
    "hotelling", # row 1
    "linear" # row 2
]

firm1 = "q"
firm2 = "dqn"

shock_none_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_a_data = {
    model: {
```

```

        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_b_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_c_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

```

```

}

for i in range(1,8):
    shock_none_df = pd.read_csv(f"../results/run_{i}/shock_none/
↪{firm1}_vs_{firm2}.csv")
    shock_a_df = pd.read_csv(f"../results/run_{i}/shock_a/
↪{firm1}_vs_{firm2}_schemeA.csv")
    shock_b_df = pd.read_csv(f"../results/run_{i}/shock_b/
↪{firm1}_vs_{firm2}_schemeB.csv")
    shock_c_df = pd.read_csv(f"../results/run_{i}/shock_c/
↪{firm1}_vs_{firm2}_schemeC.csv")

    # Map each dataframe to its corresponding data storage dictionary
    shock_mappings = [
        (shock_none_df, shock_none_data),
        (shock_a_df, shock_a_data),
        (shock_b_df, shock_b_data),
        (shock_c_df, shock_c_data)
    ]

    for df, data_dict in shock_mappings:
        for idx, model in enumerate(models):
            # Access the row matching the current model by its index (0, 1, or
↪2)

            row = df.iloc[idx]

            # Dynamically look up columns based on firm name capitalization (e.
↪g., "Q" and "PSO")
            f1_upper = firm1.upper()
            f2_upper = firm2.upper()

            # Append Average Prices
            data_dict[model]["avg_prices"][firm1].append(row[f"{f1_upper} Avg.
↪Prices"])
            data_dict[model]["avg_prices"][firm2].append(row[f"{f2_upper} Avg.
↪Prices"])

            # Append Extra-profits Δ (deltas)
            data_dict[model]["deltas"][firm1].append(row[f"{f1_upper} Δ"])
            data_dict[model]["deltas"][firm2].append(row[f"{f2_upper} Δ"])

            # Append RPD I
            data_dict[model]["rpdis"][firm1].append(row[f"{f1_upper} RPD I"])
            data_dict[model]["rpdis"][firm2].append(row[f"{f2_upper} RPD I"])

```

```
[5]: def paired_mean_diff(x, y, axis=0):
      return np.mean(x - y, axis=axis)
```

3 Logit

3.1 Shock None

3.1.1 Average Prices

```
[6]: result = permutation_test(
      data=(shock_none_data['logit']['avg_prices'][firm1],
      ↪shock_none_data['logit']['avg_prices'][firm2]),
      statistic=paired_mean_diff,
      permutation_type='samples',      # 'samples' ensures data remains paired
      alternative='two-sided',          # Options: 'two-sided', 'less', or 'greater'
      n_resamples=np.inf,               # Use np.inf for an exact test, or an
      ↪integer like 9999 for larger data
      random_state=42                  # Sets a seed for reproducibility
    )

    print("--- Paired Permutation Test Results ---")
    print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
    print(f"Exact P-value: {result.pvalue:.4f}")

    # Evaluate significance
    alpha = 0.05
    if result.pvalue <= alpha:
        print(f"Result is statistically significant (p <= {alpha}). Reject the null
        ↪hypothesis. There is a significant difference between the two firms set
        ↪prices.")
    else:
        print(f"Result is not statistically significant (p > {alpha}). Fail to
        ↪reject the null hypothesis. There is no significant difference between the
        ↪two firms set prices.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.
```

3.1.2 Deltas

```
[7]: result = permutation_test(
      data=(shock_none_data['logit']['deltas'][firm1],
      ↪shock_none_data['logit']['deltas'][firm2]),
```

```

    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    reject the null hypothesis. There is no significant difference between the
    two firms deltas.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.0543
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms deltas.

```

3.1.3 RPDIs

```

[8]: result = permutation_test(
    data=(shock_none_data['logit']['rpdis'][firm1],
    shock_none_data['logit']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

```

```

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0243

Exact P-value: 0.0156

Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

3.2 Shock A

3.2.1 Average Prices

```

[9]: result = permutation_test(
    data=(shock_a_data['logit']['avg_prices'][firm1],_
    ↪shock_a_data['logit']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms set_
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms set prices.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.

3.2.2 Deltas

```
[10]: result = permutation_test(
    data=(shock_a_data['logit']['deltas'][firm1],
    ↪shock_a_data['logit']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")
```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.1514
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms deltas.

3.2.3 RPDIs

```
[11]: result = permutation_test(
    data=(shock_a_data['logit']['rpdis'][firm1],
    ↪shock_a_data['logit']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
```

```

    n_resamples=np.inf,          # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42              # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null,
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0343
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

```

3.3 Shock B

3.3.1 Average Prices

```

[12]: result = permutation_test(
    data=(shock_b_data['logit']['avg_prices'][firm1],
    ↪shock_b_data['logit']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance

```

```

alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms set_
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms set prices.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.

```

3.3.2 Deltas

```

[13]: result = permutation_test(
    data=(shock_b_data['logit']['deltas'][firm1],_
    ↪shock_b_data['logit']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result._
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms deltas.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.0700
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.

```

There is a significant difference between the two firms deltas.

3.3.3 RPDIs

```
[14]: result = permutation_test(
    data=(shock_b_data['logit']['rpdiss'][firm1],
    ↪shock_b_data['logit']['rpdiss'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0300
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.
```

3.4 Shock C

3.4.1 Average Prices

```
[15]: result = permutation_test(
    data=(shock_c_data['logit']['avg_prices'][firm1],
    ↪shock_c_data['logit']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
```

```

    random_state=42                # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪ statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪ hypothesis. There is a significant difference between the two firms set_
    ↪ prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪ reject the null hypothesis. There is no significant difference between the_
    ↪ two firms set prices.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.

```

3.4.2 Deltas

```

[16]: result = permutation_test(
    data=(shock_c_data['logit']['deltas'][firm1],_
    ↪ shock_c_data['logit']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
    ↪ integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪ statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:

```

```

    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms deltas.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): 0.2143

Exact P-value: 0.0312

Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms deltas.

3.4.3 RPDIs

```

[17]: result = permutation_test(
    data=(shock_c_data['logit']['rpdis'][firm1],_
    ↪shock_c_data['logit']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',          # 'samples' ensures data remains paired
    alternative='two-sided',             # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,                  # Use np.inf for an exact test, or an_
    ↪integer like 9999 for larger data
    random_state=42                      # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0357

Exact P-value: 0.0156

Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

4 Hotelling

4.1 Shock None

4.1.1 Average Prices

```
[18]: result = permutation_test(
    data=(shock_none_data['hotelling']['avg_prices'][firm1],
    ↪shock_none_data['hotelling']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms set
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms set prices.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.
```

4.1.2 Deltas

```
[19]: result = permutation_test(
    data=(shock_none_data['hotelling']['deltas'][firm1],
    ↪shock_none_data['hotelling']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
```

```

    n_resamples=np.inf,          # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42              # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null,
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.0514
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms deltas.

```

4.1.3 RPDIs

```

[20]: result = permutation_test(
    data=(shock_none_data['hotelling']['rpdis'][firm1],
    ↪shock_none_data['hotelling']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:

```

```

    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0386

Exact P-value: 0.0156

Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

4.2 Shock A

4.2.1 Average Prices

```

[21]: result = permutation_test(
    data=(shock_a_data['hotelling']['avg_prices'][firm1],_
    ↪shock_a_data['hotelling']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms set_
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms set prices.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0029

Exact P-value: 0.5000

Result is not statistically significant (p > 0.05). Fail to reject the null

hypothesis. There is no significant difference between the two firms set prices.

4.2.2 Deltas

```
[22]: result = permutation_test(
    data=(shock_a_data['hotelling']['deltas'][firm1],
    ↪shock_a_data['hotelling']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.0157
Exact P-value: 0.2500
Result is not statistically significant (p > 0.05). Fail to reject the null
hypothesis. There is no significant difference between the two firms deltas.
```

4.2.3 RPDIs

```
[23]: result = permutation_test(
    data=(shock_a_data['hotelling']['rpdis'][firm1],
    ↪shock_a_data['hotelling']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)
```

```

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
↳reject the null hypothesis. There is no significant difference between the_
↳two firms RPDIs.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0300
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

```

4.3 Shock B

4.3.1 Average Prices

```

[24]: result = permutation_test(
    data=(shock_b_data['hotelling']['avg_prices'][firm1],_
↳shock_b_data['hotelling']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
↳integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms set_
↳prices.")

```

```

else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms set prices.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0100

Exact P-value: 0.0156

Result is statistically significant (p ≤ 0.05). Reject the null hypothesis.

There is a significant difference between the two firms set prices.

4.3.2 Deltas

```

[25]: result = permutation_test(
    data=(shock_b_data['hotelling']['deltas'][firm1],
    ↪shock_b_data['hotelling']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): 0.0457

Exact P-value: 0.0469

Result is statistically significant (p ≤ 0.05). Reject the null hypothesis.

There is a significant difference between the two firms deltas.

4.3.3 RPDIs

```
[26]: result = permutation_test(
    data=(shock_b_data['hotelling']['rpdis'][firm1],
    ↪shock_b_data['hotelling']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0386
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.
```

4.4 Shock C

4.4.1 Average Prices

```
[27]: result = permutation_test(
    data=(shock_c_data['hotelling']['avg_prices'][firm1],
    ↪shock_c_data['hotelling']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)
```

```

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms set_
↳prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
↳reject the null hypothesis. There is no significant difference between the_
↳two firms set prices.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0086
Exact P-value: 0.0312
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms set prices.

```

4.4.2 Deltas

```

[28]: result = permutation_test(
    data=(shock_c_data['hotelling']['deltas'][firm1],_
↳shock_c_data['hotelling']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
↳integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms deltas.")
else:

```

```

    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0157

Exact P-value: 0.5938

Result is not statistically significant (p > 0.05). Fail to reject the null hypothesis. There is no significant difference between the two firms deltas.

4.4.3 RPDIs

```

[29]: result = permutation_test(
    data=(shock_c_data['hotelling']['rpdis'][firm1],
    ↪shock_c_data['hotelling']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0257

Exact P-value: 0.0156

Result is statistically significant (p <= 0.05). Reject the null hypothesis. There is a significant difference between the two firms RPDIs.

5 Linear

5.1 Shock None

5.1.1 Average Prices

```
[30]: result = permutation_test(
    data=(shock_none_data['linear']['avg_prices'][firm1],
    ↪shock_none_data['linear']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms set
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms set prices.")
```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0086

Exact P-value: 0.0312

Result is statistically significant (p <= 0.05). Reject the null hypothesis.

There is a significant difference between the two firms set prices.

5.1.2 Deltas

```
[31]: result = permutation_test(
    data=(shock_none_data['linear']['deltas'][firm1],
    ↪shock_none_data['linear']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
```

```

    n_resamples=np.inf,          # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42              # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): 0.0000
Exact P-value: 1.0000
Result is not statistically significant (p > 0.05). Fail to reject the null
hypothesis. There is no significant difference between the two firms deltas.

```

5.1.3 RPDIs

```

[32]: result = permutation_test(
    data=(shock_none_data['linear']['rpdis'][firm1],
    ↪shock_none_data['linear']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:

```

```

    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0186

Exact P-value: 0.0156

Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

5.2 Shock A

5.2.1 Average Prices

```

[33]: result = permutation_test(
    data=(shock_a_data['linear']['avg_prices'][firm1],_
    ↪shock_a_data['linear']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an_
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
    ↪hypothesis. There is a significant difference between the two firms set_
    ↪prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
    ↪reject the null hypothesis. There is no significant difference between the_
    ↪two firms set prices.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0071

Exact P-value: 0.0625

Result is not statistically significant (p > 0.05). Fail to reject the null

hypothesis. There is no significant difference between the two firms set prices.

5.2.2 Deltas

```
[34]: result = permutation_test(
    data=(shock_a_data['linear']['deltas'][firm1],
    ↪shock_a_data['linear']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0186
Exact P-value: 0.5000
Result is not statistically significant (p > 0.05). Fail to reject the null
hypothesis. There is no significant difference between the two firms deltas.
```

5.2.3 RPDIs

```
[35]: result = permutation_test(
    data=(shock_a_data['linear']['rpdis'][firm1],
    ↪shock_a_data['linear']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)
```

```

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
↳reject the null hypothesis. There is no significant difference between the_
↳two firms RPDIs.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0100
Exact P-value: 0.0312
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.

```

5.3 Shock B

5.3.1 Average Prices

```

[36]: result = permutation_test(
    data=(shock_b_data['linear']['avg_prices'][firm1],_
↳shock_b_data['linear']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an_
↳integer like 9999 for larger data
    random_state=42                 # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms set_
↳prices.")

```

```

else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms set prices.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0086

Exact P-value: 0.0312

Result is statistically significant (p <= 0.05). Reject the null hypothesis.

There is a significant difference between the two firms set prices.

5.3.2 Deltas

```

[37]: result = permutation_test(
    data=(shock_b_data['linear']['deltas'][firm1],
    ↪shock_b_data['linear']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}) : {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms deltas.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): 0.0357

Exact P-value: 0.0312

Result is statistically significant (p <= 0.05). Reject the null hypothesis.

There is a significant difference between the two firms deltas.

5.3.3 RPDIs

```
[38]: result = permutation_test(
    data=(shock_b_data['linear']['rpdis'][firm1],
    ↪shock_b_data['linear']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")
```

```
--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0157
Exact P-value: 0.0156
Result is statistically significant (p <= 0.05). Reject the null hypothesis.
There is a significant difference between the two firms RPDIs.
```

5.4 Shock C

5.4.1 Average Prices

```
[39]: result = permutation_test(
    data=(shock_c_data['linear']['avg_prices'][firm1],
    ↪shock_c_data['linear']['avg_prices'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)
```

```

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms set_
↳prices.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to_
↳reject the null hypothesis. There is no significant difference between the_
↳two firms set prices.")

```

```

--- Paired Permutation Test Results ---
Observed Mean Difference (Q - DQN): -0.0043
Exact P-value: 0.2500
Result is not statistically significant (p > 0.05). Fail to reject the null
hypothesis. There is no significant difference between the two firms set prices.

```

5.4.2 Deltas

```

[40]: result = permutation_test(
    data=(shock_c_data['linear']['deltas'][firm1],_
↳shock_c_data['linear']['deltas'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,              # Use np.inf for an exact test, or an_
↳integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
↳statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null_
↳hypothesis. There is a significant difference between the two firms deltas.")
else:

```

```

    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms deltas.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): 0.1200

Exact P-value: 0.1562

Result is not statistically significant (p > 0.05). Fail to reject the null hypothesis. There is no significant difference between the two firms deltas.

5.4.3 RPDIs

```

[41]: result = permutation_test(
    data=(shock_c_data['linear']['rpdis'][firm1],
    ↪shock_c_data['linear']['rpdis'][firm2]),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' ensures data remains paired
    alternative='two-sided',         # Options: 'two-sided', 'less', or 'greater'
    n_resamples=np.inf,             # Use np.inf for an exact test, or an
    ↪integer like 9999 for larger data
    random_state=42                  # Sets a seed for reproducibility
)

print("--- Paired Permutation Test Results ---")
print(f"Observed Mean Difference ({firm1.upper()} - {firm2.upper()}): {result.
    ↪statistic:.4f}")
print(f"Exact P-value: {result.pvalue:.4f}")

# Evaluate significance
alpha = 0.05
if result.pvalue <= alpha:
    print(f"Result is statistically significant (p <= {alpha}). Reject the null
    ↪hypothesis. There is a significant difference between the two firms RPDIs.")
else:
    print(f"Result is not statistically significant (p > {alpha}). Fail to
    ↪reject the null hypothesis. There is no significant difference between the
    ↪two firms RPDIs.")

```

--- Paired Permutation Test Results ---

Observed Mean Difference (Q - DQN): -0.0143

Exact P-value: 0.0312

Result is statistically significant (p <= 0.05). Reject the null hypothesis. There is a significant difference between the two firms RPDIs.